(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

EIGHTH SEMESTER [B.TECH] JUNE 2024

Paper Code: ETIT-402 ***Subject: Mobile Computing***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data if any.

Q1 Attempt **any Five** of the following questions:- **[5x5=25]**

a) What were the reason behind the implementation of WAP services with WML?
b) Mention the features of mobile internet.
c) Explain the WAP architecture in brief.
d) Explain the Data dissemination issues in mobile networks.
e) Differentiate between tunneling and reverse tunneling.
f) With the help of neat protocol stack, draw and explain three-tier architecture for mobile computing. What are mobile nodes? Explain.

UNIT–I

Q2 What is mobile computing? List its key characteristics. Write down the applications of mobile computing. **[12.5]**

Q3 a) Explain Generation of Mobile services. **[6]**
b) Describe the working of a contention-based MAC protocol with suitable examples. **[6.5]**

UNIT–II

Q4 Compare and Explain IEEE 802.11 (WLAN) and IEEE 802.15 (Bluetooth). **[12.5]**

Q5 Write down Short note on following (**any two**): **[6.25, 6.25]**

a) WiMax technology and its application.
b) ZigBee Technology and its usage.
c) Mobile Handover Procedure

UNIT–III

Q6 a) Differentiate between ad-hoc network and infrastructure-based network. **[6]**
b) Explain DSR routing protocol in details. **[6.5]**

Q7 a) What is snooping TCP approach in mobile wireless networks? Discuss its advantages. **[6]**
b) Explain location management in mobile networks. **[6.5]**

UNIT–IV

Q8 a) What are the various methods for Data Synchronization? Explain. **[6]**
b) Explain the concept of wireless markup language. **[6.5]**

Q9 a) Compare the feature sets of Android, Symbian, and Windows Phone operating systems. **[6]**
b) Explain the architecture palm OS. What are the constraints of mobile device OS? **[6.5]**

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ETIT-402

(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Eighth Semester, April, 2024

Paper Code: ETIT-402 **Subject: Mobile Computing**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks
1	Write short notes on the following: - (a) Define MSC-SA and PLNM. (b) Explain Licensed and Unlicensed Bands. (c) Define IMSI and IMEI. (d) Why Uplink frequencies are always selected as the lower range of band than downlink frequencies.	(2.5*4)
2	(a) Explain hidden Terminal Problem. How it is controlled in IEEE 802.11? (b) Discuss Piconet, Scatternet and Bluetooth architecture specifications.	(5*2)
3	(a) Explain the architecture of GPRS showing all the interfaces. (b) Explain various capacity expansion techniques with diagrams (Cell Splitting and Cell Sectoring).	(5*2)
4	Write short notes on the following: (a) ZigBee Technology (b) IrDA	(5*2)

(Please write your Roll No. immediately) Roll No. .....

MID TERM EXAMINATION

EIGHTH SEMESTER (B. TECH), April 2024

Paper code: ETIT-402 Subject: Mobile Computing
Time: 1 Hour 30 Minutes Maximum Marks: 30

Note: Attempt three questions only.

1. **Question No.1 is compulsory.**
2. **Attempt any two questions from the remaining three questions.**

Q-1. Explain each of the following in short answer: (5 x 2)

(a) How do hard handoff and soft handoff differ in mobile cellular networks?

(b) Outline the characteristics of different generations of cellular systems?

(c) Using diagrams, differentiate between hidden terminals and exposed terminals in WLAN systems.

(d) Enumerate the fundamental features of a CDMA system.

(e) What are the primary functions of the Home Location Register (HLR) and Visitor Location Register (VLR) in mobile networks?

Q-2. (a) Evaluate the effectiveness of the three-tier architecture in mobile computing for ensuring scalability and flexibility in mobile applications. Provide examples to support your analysis. (5)

(b) Critically assess the role of WAP (Wireless Application Protocol) in facilitating mobile application development and deployment. Discuss its strengths and limitations, considering factors such as security, protocol stack, and application environment. (5)

Q-3.

(a) Evaluate the suitability of Bluetooth technology for various user scenarios outlined in the IEEE 802.15 standard. Consider factors such as data transfer rates, power consumption, and compatibility with different devices. (5)

(b)Compare and contrast the multiple access techniques used in wireless LANs, such as collision avoidance, polling, and spread spectrum. (5)

Q-4. Write a short note on **any two** of the following. (4 x 2.5)

a) What are some common user scenarios for Bluetooth as defined by the IEEE 802.15 standard?
b) Describe the physical layer specifications for Bluetooth technology.
c) What functionalities are provided by the MAC layer in Bluetooth communication?
d) How is link management handled in Bluetooth networks?

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(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

EIGHTH SEMESTER [B.TECH] JULY 2023

Paper Code: ETIT-402 ***Subject: Mobile Computing***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q. No. 1 which is compulsory.

Q1 Answer in brief **(Any five)**:- **(5x5=25)**

(a) Compare Fixed Channel Assignment and Dynamic Channel Assignment and Hybrid Channel Assignment (HCA).
(b) Explain the Handoff process.
(c) What is Near and Far terminal and Hidden-Exposed Problem?
(d) What are the two different basic transmission technologies used to set up WLAN?
(e) What are the various problems which are related to the use of DHCP? How they are overcome?
(f) Discuss the challenges and problems of a handling file system in mobile environment.
(g) What are the advantages of data hoarding? Explain M, E, S, and I tags in detail.

Q2 (a) Draw three tier architecture for mobile computing. Explain. **(6)**
(b) Compare CDMA with GPRS. **(6.5)**

Q3 (a) Assume all stations can hear all other stations. One station wants to transmit and senses the carrier idle. Why can a collision still occur after the start of transmission? **(6)**
(b) Explain the architecture of Bluetooth technology with the help of neat diagram. **(6.5)**

Q4 (a) Differentiate between Dynamic source routing and Ad Hoc on demand distance vector routing. **(6)**
(b) What are the goals of mobile IP? How the packet delivery mechanism works for mobile IP? **(6.5)**

Q5 Compare the following technologies in detail:- **(5x2.5=12.5)**

(a) WPABX
(b) IrDA
(c) ZigBee
(d) RFID
(e) WiMax

Q6 (a) Explain the process of caching invalidation? Discuss its solutions. **(6)**
(b) Mention the features of HIPER-LAN and compare it with 802.11a, 802.11b, 802.11g. **(6.5)**

Q7 (a) Discuss how ad-hoc network differs from wired network in the aspect of routing? Explain any two on demand routing protocol used for routing in this network. **(6)**
(b) Explain the working procedure of snooping TCP using neat diagram. Elaborate the advantages and disadvantages of Snooping TCP. **(6.5)**

Q8 (a) Give an overview of the WAP architecture and discuss its protocols and components. Compares this architecture with the typical internet architecture. **(6)**
(b) What are the main challenges of VoIP? How IPSEC helps in making VoIP network as secure? Discuss. **(6.5)**

Q9 (a) Explain the i-mode service. How wireless session protocol (WSP) operate on top of the datagram service WDP. **(6)**
(b) Explain Palm OS. Discuss main features of Palm OS in detail What are the advantages of Palm operating system? **(6.5)**

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ETIT-402

MID TERM EXAMINATION

EIGHTH SEMESTER [B. TECH.] APRIL 2022

Paper Code: ETIT402 **Subject: Mobile Computing**

Time: 1Hour 30 Min. **Maximum Marks: 30**

Note: Attempt any Three questions including Q. No. 1 which is compulsory.

Q1. **(a)** **Explain the Mobile Computing Design consideration in detail.** **(2)**

(b) **What do you mean by handoffs? Explain its types** **(2)**

(c) **What do you understand by mobile database?** **(2)**

(d) **Explain the following terms:** **(4)**

i) **WiMax**

ii) **Issues in Mobile Computing**

Q2. **(a)** **How location management is done in GSM architecture using HLR-VLR.** **(8)**

(b) **Explain the term CSMA. How it is helpful in detecting collision.** **(2)**

Q3. **(a)** **Show the diagrammatic representation of the multiple access techniques using FDMA, TDMA, CDMA in detail. Compare TDMA and FDMA schemes.** **(4)**

(b) **Describes any 12 terminology used in Mobile Telephony in detail.** **(6)**

Q4. **(a)** **Explain the 3-tier architecture of mobile computing.** **(6)**

(b) **Discuss the Layered architecture of WAP model. Explain the features of WTLS.** **(4)**

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END TERM EXAMINATION

EIGHTH SEMESTER [B.TECH.]

MID TERM EXAM

Paper Code:2 *Subject: MOBILE COMPUTING*

Time:1hr 30 *Maximum Marks: 30*

Note three questions including Q. no. 1 which is compulsory.

Q1.(a)Explile telephony (2x5=10)

(b)What dan by ISMA techniques.

(c) Explaind soft hndsoff technique.

(d) ExplainModule

(e)Explain Lan protocol

Q2(a)Explatechnique with an example (5)

(b)Explain rence between TDMA and FDMA techniques.(5)

Q3(a)Explaarchitecture .(5)

(b)Explain access control 5)

Q4(a)Explarchitecture of Bluetooth. (5)

(b)What is concept and define the role of frequency .(5)